

DMRG class at BME – lecture plan

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Projects

You may choose **one** of the projects below.

1. Symmetry Protected Topological order in the anisotropic Heisenberg Chain

Consider the $S = 1$ Heisenberg chain with uniaxial anisotropy,

$$H = \sum_i \left(J \vec{S}_i \cdot \vec{S}_{i+1} + D(S_i^z)^2 \right) ,$$

where $S_i^{x,y,z}$ are the 3×3 spin matrices of an $S = 1$ spin.

- Construct the MPO form of the Hamiltonian.
- Calculate the ground state for various anisotropy strengths $D/J \in [0, 2.5]$, using the infinite-chain (chain growing) DMRG algorithm. Approximate the energy density ($\epsilon = E/L$) in the $L \rightarrow \infty$ limit? (*Hint: instead of dividing the energy by the actual length, it is better to calculate the 'bond-energy' $\epsilon = \langle J \vec{S}_i \cdot \vec{S}_{i+1} + D/2(S_i^z)^2 + D/2(S_{i+1}^z)^2 \rangle$*)
- From the Schmidt values, calculate the von Neumann entropy between the two half systems, and plot the $S(D/J)$ function.
- Consider the Schmidt-spectrum of the ground state. What do you observe for small D/J ? And for large D/J ? Where is the transition point?
- Investigate, how your results depend on the MPS bond dimension M .
- (EXTRA TASK!) You may also calculate the energy gap by requiring eigs() to produce 2 eigenstates. (*Hint: if calculating 2 eigenstates simultaneously in DMRG, it is better to diagonalize the mixture of the block reduced density matrices, i.e. $\rho_{\text{left}} = 0.5\rho_{\text{left}}^{\text{GS}} + 0.5\rho_{\text{left}}^{\text{exc}}$ to form better block states both in the left and the right.*)

2. Ground state of the $S = 1$ bilinear-biquadratic model

Consider the $S = 1$ "bilinear-biquadratic" model,

$$H = \sum_i \left(\cos(\theta) \vec{S}_i \cdot \vec{S}_{i+1} + \sin(\theta) (\vec{S}_i \cdot \vec{S}_{i+1})^2 \right) ,$$

where $\vec{S}_i = (S_i^x, S_i^y, S_i^z)$ and $S_i^{x,y,z}$ are the 3×3 spin matrices of an $S = 1$ spin.

- Construct the MPO form of the Hamiltonian. What is the bond-dimension of the MPO?
- Calculate the ground state as a function of $\theta \in [-\pi/2, \pi/2]$ using the infinite chain (chain-growing) DMRG algorithm. Approximate the energy density ($\epsilon = E/L$) in the $L \rightarrow \infty$ limit? (*Hint: instead of dividing the energy by the actual length, it is better to calculate the 'bond-energy' $\epsilon = \langle \cos(\theta) \vec{S}_i \cdot \vec{S}_{i+1} + \sin(\theta) (\vec{S}_i \cdot \vec{S}_{i+1})^2 \rangle$*)

- From the Schmidt values, calculate the von Neumann entropy between the two half systems, and plot the $S(\theta)$ function.
- Consider the following points: $\theta = \{0, \arctan(1/3), \pm\pi/4, -\pi/2\}$ How do the entanglement entropy depend on the system size L in these points? Calculate also the $\langle \vec{S}_i \cdot \vec{S}_{i+1} \rangle$ nearest neighbor correlator at the two central sites as a function of L . What did you learn about the ground state?
- Investigate, how your results depend on the MPS bond dimension M .

3. Information scrambling, out-of-time-ordered correlators (OTOC's)

Consider the mixed-field Ising chain,

$$H = - \sum_i (J Z_i Z_{i+1} + g Z_i + h X_i)$$

with $J = 1$, $h = -1.05$, and $g = 0.5$, where Z_i and X_i are the usual spin-1/2 spin operators (Pauli matrices). Consider a chain of length $L \approx 20 - 40$. The goal of this project is to calculate the so-called out-of-time-ordered correlator (OTOC) of the Z operators,

$$C(r, t) = \langle \Psi | [Z_{L/2}(t), Z_{L/2+r}(0)]^\dagger [Z_{L/2}(t), Z_{L/2+r}(0)] | \Psi \rangle ,$$

where $Z_i(t) = e^{iHt} Z_i e^{-iHt}$ is the time-dependent operator in the Heisenberg picture, $[A, B] = AB - BA$ is the commutator, and $|\Psi\rangle$ is a random product state in the Z -basis, i.e., $|\Psi\rangle \sim |\uparrow\uparrow\downarrow\uparrow\downarrow \dots \downarrow\downarrow\rangle$ for some random spin configuration $\{\sigma_i\}$.

Note that $C(r, t) = (\langle \Psi_{12}^r(t) | - \langle \Psi_{21}^r(t) |) (| \Psi_{12}^r(t) \rangle - | \Psi_{21}^r(t) \rangle)$, with

$$| \Psi_{12}^r(t) \rangle = e^{iHt} Z_{L/2} e^{-iHt} Z_{L/2+r} | \Psi \rangle ,$$

$$| \Psi_{21}^r(t) \rangle = Z_{L/2+r} e^{iHt} Z_{L/2} e^{-iHt} | \Psi \rangle .$$

In the "12" case, the action by $Z_{L/2+r}$ is trivial as the initial state is an eigenstate of $Z_{L/2+r}$, i.e., we only get a ± 1 prefactor depending on r and the initial spin configuration. Therefore, in both cases, first we need to evolve the state $|\Psi\rangle$ to t , then act by $Z_{L/2}$, then evolve back in time (with $-dt$ steps) to 0. In case of the "12" branch, we then only need to apply the ± 1 factor depending on r and the initial configuration, while in case of "21" we need to apply $Z_{L/2+r}$ on the state.

- Construct TEBD on the finite chain of length L with open boundary conditions.
- Use time-steps $dt \approx 0.05 - 0.1$ and bond dimensions $M \lesssim 100 - 200$ depending on your machine. (run the calculations at 2-3 different M values)
- Determine $C(r, t)$ up to times $T_{\max} \approx 10$, and plot it on a heatmap.

4. Dynamical Phase Transitions, Loschmidt-echo

Consider the transverse field Ising model,

$$H = - \sum_i (J Z_i Z_{i+1} + h X_i) .$$

In this model, a quantum phase transition (QPT) occurs at $h/J = 1$. We initialize the system in the ground state at $H(h_0)$. At $t = 0$ we change the transverse field to h_1 (sudden global quench), leading to a time-dependent quantum state $|\Psi(t)\rangle$. The Loschmidt-Echo rate function in the thermodynamic limit is defined as

$$\lambda(t) = - \lim_{L \rightarrow \infty} \frac{1}{L} \ln |\langle \Psi(0) | \Psi(t) \rangle|^2 .$$

Our goal is to extract this function by simulating the dynamics using the infinite-chain TEBD (iTEBD) algorithm.

The initial field is set to $h_0 = 0$; therefore, the initial state is a product (Neél state: $\dots \uparrow\downarrow\uparrow\downarrow\dots$).

Note that in iTEBD, you have only two matrices (A & B) in the MPS, and $\lambda(t)$ is related to the largest magnitude eigenvalue of the transfer matrix

$$T_{\alpha\tilde{\alpha}\beta\tilde{\beta}} = \sum_{\sigma_1, \sigma_2} \sum_{\gamma, \tilde{\gamma}} A(t)_{\alpha\gamma}^{\sigma_1} B(t)_{\gamma\beta}^{\sigma_2} \left(A(0)_{\tilde{\alpha}\tilde{\gamma}}^{\sigma_1} \right)^* \left(B(0)_{\tilde{\gamma}\tilde{\beta}}^{\sigma_2} \right)^* .$$

As the initial state is a product, $\dim(\tilde{\alpha}) = \dim(\tilde{\beta}) = \dim(\tilde{\gamma}) = 1$, i.e., T is an $M \times M$ matrix, whose largest magnitude eigenvalue at time t is $\tau(t)$. The overlap of a long segment of length L is then $\tau^{L/2}$, therefore,

$$\lambda(t) = - \lim_{L \rightarrow \infty} \frac{1}{L} \ln |\tau(t)|^L = - \ln |\tau(t)|$$

- Set up an iTEBD calculation with the Neél state as an initial state.
- Evolve the state for an external field h_1 , and at every timestep t , form the T transfer matrix.
- By the diagonalization of the T transfer matrix, determine the Loschmidt echo rate function $\lambda(t)$.
- Plot and analyze $\lambda(t)$. How does it look like, if $h_1 < 1$? And if $h_1 > 1$?
- Perform your calculations for bond dimensions $M \lesssim 200$ depending on your computer. How do your results depend on M ?
- Plot also the half-chain entropy as a function of t for different M 's. Up to what times are your results reliable?

Literature tasks

If you decide to get a grade by reading literature, select **one** of the following advanced topics, read the highlighted papers. On the exam, you are supposed to present the content. (You may take notes/perform calculations, that you can use at the exam).

1. Projected Entangled Pair States

- F. Verstraete and J. I. Cirac, "Renormalization algorithms for Quantum-Many Body Systems in two and higher dimensions", arXiv:0407066 (2004).
- R. Orús, "A practical introduction to tensor networks: Matrix product states and projected entangled pair states", Annals of Physics 349, 117–158 (2014).

2. Multi-scale Entanglement Renormalization Ansatz

- G. Vidal, "Class of Quantum Many-Body States That Can Be Efficiently Simulated", Phys. Rev. Lett. 101, 110501 (2008).
- G. Evenbly and G. Vidal, "Algorithms for entanglement renormalization", Phys. Rev. B 79, 144108 (2009).

3. Non-Abelian symmetries in TEBD

- M. A. Werner, C. P. Moca, Ö. Legeza, M. Kormos, and G. Zaránd, "Spin fluctuations after quantum quenches in the $S = 1$ Haldane chain: Numerical validation of the semi-semiclassical theory", Phys. Rev. B 100, 035401 (2019).

- M. A. Werner, C. P. Moca, Ö. Legeza, and G. Zaránd, "Quantum quench and charge oscillations in the SU(3) Hubbard model: A test of time evolving block decimation with general non-Abelian symmetries", Phys. Rev. B 102, 155108 (2020).

4. Controlled bond expansion

- Andreas Gleis, Jheng-Wei Li, and Jan von Delft, "Controlled Bond Expansion for Density Matrix Renormalization Group Ground State Search at Single-Site Costs", Phys. Rev. Lett. 130, 246402 (2023).
- Jheng-Wei Li, Andreas Gleis, and Jan von Delft, "Time-Dependent Variational Principle with Controlled Bond Expansion for Matrix Product States", Phys. Rev. Lett. 133, 026401 (2024)